15 experiment

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clc; clear;

close all;

% Traffic data (in Mbps) traffic = [120 150 180 210

250 280 320 350 390 420];

% Time values

time = 1:length(traffic);

% Linear regression for traffic prediction p

= polyfit(time, traffic, 1);

% Future time future_time

= 11:15;

% Predicted traffic predicted_traffic =

polyval(p, future_time);

% Display results disp('Future Traffic

Prediction (Mbps):');

disp(predicted_traffic');

% Plot actual traffic plot(time, traffic,

'o-', 'LineWidth', 2); hold on;

% Plot predicted traffic plot(future_time,

predicted_traffic, 'x--', 'LineWidth', 2); xlabel('Time');

ylabel('Traffic (Mbps)'); title('AI-Based Traffic Prediction

for 5G Networks'); legend('Actual Traffic', 'Predicted

Traffic'); grid on;